1. Practical5b.c

```
    #include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
  int fd[2];

  pipe(fd);

  if(fork()==0)
  {
    dup2(fd[1], STDOUT_FILENO);
    close(fd[0]);
    close(fd[1]);
    execlp("ls","ls","-l",NULL);
  }

  if(fork()==0)
  {
    dup2(fd[0], STDIN_FILENO);
    close(fd[0]);
    close(fd[1]);
    execlp("grep","grep",".c",NULL);
  }
close(fd[0]);
  close(fd[1]);

  wait(NULL);
  wait(NULL);

  return 0;
}
```

```
hamsini@HamsiniVanam:~ $ gcc Practical5b.c -o Practical5b
hamsini@HamsiniVanam:~ $ ls -l
total 24
-rwxr-xr-x 1 root root 16224 Sep 29 05:15 Practical5b
-rw-r--r-- 1 root root   491 Sep 29 05:15 Practical5b.c
-rw-r--r-- 1 root root     0 Sep 29 05:15 a.c
-rw-r--r-- 1 root root     0 Sep 29 05:15 b.c
-rw-r--r-- 1 root root     0 Sep 29 05:15 notes.txt
-rw-r--r-- 1 root root   462 Sep 29 05:15 signal_demo.c
hamsini@HamsiniVanam:~ $ ./Practical5b
-rwxr-xr-x 1 root root 16224 Sep 29 05:15 Practical5b
-rw-r--r-- 1 root root   491 Sep 29 05:15 Practical5b.c
-rw-r--r-- 1 root root     0 Sep 29 05:15 a.c
-rw-r--r-- 1 root root     0 Sep 29 05:15 b.c
-rw-r--r-- 1 root root   462 Sep 29 05:15 signal_demo.c
```

2. Create a POSIX signal handling program that captures SIGINT, SIGTERM, and SIGUSR1. Demonstrate asynchronous event handling and explain the role of signal handlers.

```
#include <stdio.h>
#include <signal.h>
#include <unistd.h>

void handler(int sig)
{
   if(sig==SIGINT)
      printf("\nSIGINT Received\n");

   if(sig==SIGTERM)
      printf("\nSIGTERM Received\n");

   if(sig==SIGUSR1)
      printf("\nSIGUSR1 Received\n");
}

int main()
{
   signal(SIGINT,handler);
   signal(SIGTERM,handler);
   signal(SIGUSR1,handler);

   while(1)
   {
      printf("Running...\n");
      sleep(3);
   }

   return 0;
}
```

```
hamsini@HamsiniVanam:~ $ ps -e | grep signal_demo
    156 pts/0    00:00:00 signal_demo
hamsini@HamsiniVanam:~ $ kill -SIGUSR1 156
hamsini@HamsiniVanam:~ $ kill -SIGTERM 156
hamsini@HamsiniVanam:~ $ kill -SIGINT 156
hamsini@HamsiniVanam:~ $ kill -9 156
```